

JomExam

ESAT Biology — Sample Paper

Six questions in the style of ESAT Biology, including a graph, a paired-options question and a statement-set question. Allow 9 minutes — the real paper's 90-second pace. No calculator. One option is correct in each question, and there is no negative marking.

Worked solutions with the trap named follow the questions. The full experience — 27 questions, 40 minutes, and a report naming the skill behind every wrong answer — is free at jomexam.com.

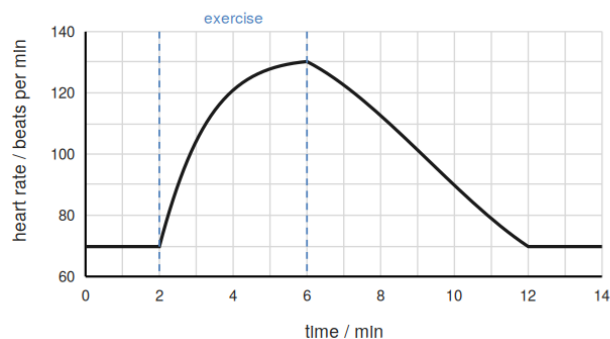
Question 1. A human body cell contains 46 chromosomes. How many chromosomes are there in each cell produced when a cell in the testes divides by MEIOSIS?

- A 46
- B 92
- C 12
- D 23
- E 44

Question 2. A field has an area of 600 m^2 . A student places twenty 0.25 m^2 quadrats at random and counts a mean of 3 daisy plants per quadrat. What is the estimate of the daisy population of the field?

- A 60
- B 1 800
- C 7 200
- D 240
- E 36 000

Question 3. The graph shows a student's heart rate before, during and after a period of exercise. How long after the exercise ENDS does the heart rate return to its resting value?



- A 4 min
- B 6 min
- C 8 min
- D 12 min
- E 2 min

Question 4. Which of the following best describes negative feedback in the control of body temperature?

- A A change in temperature triggers responses that amplify the change.

- B** A change in temperature triggers responses that reverse the change, returning the body toward the set level.
- C** The body prevents its temperature from ever changing.
- D** Sweating and shivering occur together to hold the temperature steady.
- E** The brain switches temperature control off once the set level is reached.

Question 5. Blood glucose is regulated by hormones. Which row correctly pairs a hormone with its effect?

	Hormone	Effect
A	insulin	causes the liver to convert glycogen into glucose
B	glucagon	causes the liver to convert glucose into glycogen
C	insulin	released when blood glucose falls too low
D	glucagon	causes the liver to convert glycogen into glucose
E	adrenaline	lowers the heart rate during exercise
F	insulin	causes muscle cells to release stored glucose
G	glucagon	released by the liver when glucose rises
H	adrenaline	causes glucose to be stored as glycogen

Question 6. A population of bacteria is treated with an antibiotic, and over several years resistant strains become common. Consider the following statements:

1. Mutations giving resistance arise randomly, before the bacteria meet the antibiotic.
2. The antibiotic causes the bacteria to mutate so that they become resistant.
3. Resistant bacteria survive the treatment and pass the resistance to their offspring.

Which statements are correct?

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

Worked solutions

Question 1 — answer D.

1. Meiosis halves the chromosome number: two divisions produce four haploid cells.
2. $46/2 = 23$.

The trap: The trap is 46 — mitosis's answer given to meiosis's question. Gametes must be haploid so that fertilisation restores 46; a 46-chromosome sperm would build a 92-chromosome embryo.

Question 2 — answer C.

1. Mean density: 3 plants per $0.25 \text{ m}^2 = 12$ plants per m^2 .
2. Population $\approx 12 \times 600 = 7\,200$.

The trap: The trap is 1 800 — multiplying 3 by 600 as if each square metre held one quadrat's count. The quadrat is a QUARTER of a square metre; the denominator is where sampling questions collect their marks.

Question 3 — answer B.

1. Exercise ends at $t = 6$ min (the dashed line).
2. The curve returns to the resting 70 beats per minute at $t = 12$ min: $12 - 6 = 6$ minutes.

The trap: The trap is 12 — the clock time read as the recovery time. The question asks for time AFTER the exercise ends, and the subtraction is the entire mark.

Question 4 — answer B.

1. Negative feedback detects a change and drives the OPPOSITE response, returning the variable toward the set level.

The trap: Option C is the near-miss: feedback does not PREVENT change — it responds to change that has already happened. The correction, not the constancy, is the mechanism.

Question 5 — answer D.

1. Glucagon is released when glucose falls, and instructs the liver to convert stored glycogen back into glucose.

The trap: Rows A and B are the two halves of the classic swap: insulin and glucagon with each other's jobs. The pairing, not the vocabulary, is what the question checks.

Question 6 — answer F.

1. Statement 1: resistance mutations arise randomly, before exposure — true.
2. Statement 2: the antibiotic SELECTS, it does not instruct — false, and the single most examined misconception in this topic.
3. Statement 3: survival and reproduction of the resistant — true. So 1 and 3 only.

The trap: Statement 2 is the heart of the question: mutation is random, selection is not. Any answer accepting it has natural selection running backwards, with the environment writing the mutations it prefers.